ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · A7 · ENVIRONMENTS

Habitats and food webs

★☆☆
Grade 5

Learning goals

- Use the key vocabulary.
- Read a food web.

1 Match each word to its meaning. [6]

Draw a line to match each pair, or write the letter.

producer	→ F	**A**	eats only animals
herbivore	→ E	**B**	eats plants and animals
carnivore	→ A	**C**	the place where an organism lives
omnivore	→ B	**D**	breaks down dead plants and animals
decomposer	→ D	**E**	eats only plants
habitat	→ C	**F**	makes its own food using light

2 Use the food web on the "Learn it" page to answer. [4]

a) Name the two producers.

mangrove leaves and algae

b) Write one food chain with four links.

algae → shrimp → fish → kingfisher (or algae → fish → otter, etc.)

c) Name a top predator.

tiger (or kingfisher / otter)

d) What do crabs eat?

mangrove leaves

3 **True or false?** [4]

Statement	True	False	If false, correct it
1. Every food chain starts with a producer.	✔		
2. The arrows in a food web point from the eater to the food.		✔	They point from the food to the eater (direction of energy).
3. Decomposers return nutrients to the soil.	✔		
4. A field with one crop has high biodiversity.		✔	It has low biodiversity.

4 **Fill in the gaps.** [4]

WORD BANK

biomimicry **sunlight** **consumers** **soil** **rain** **food**

Energy enters a food web as sunlight. Plants use it to make food. Energy passes to consumers when they eat. Copying ideas from nature is called biomimicry.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · A7 · ENVIRONMENTS

What happens if...?

★★☆
Grade 5

Learning goals

- Predict changes in a food web.
- Explain energy flow.
- Link features of living things to designs.

1 **Use the Sundarbans food web. Predict what would happen and explain why.** [4]

a) If many deer died from disease, what would happen to the tigers?

Tigers would have less food, so their numbers might fall, or they would eat more monkeys.

b) If an oil spill killed the algae, which animals would be affected? Explain.

Shrimp and fish (which eat algae) would have less food, then kingfishers and otters that eat them would be affected too.

2 **Why are there far fewer tigers than deer in the Sundarbans?** [2]

Energy is lost at each step of the food chain (used for moving and keeping warm), so there is only enough energy for a few top predators.

3 **Complete the table about biomimicry.** [6]

Feature in nature	What it does	Human invention
burr hooks	catch on fur to spread seeds	Velcro
kingfisher beak	enters water smoothly with little splash	bullet train nose
termite mound	keeps the nest cool with air channels	naturally cooled buildings

4 **Design a solution by mimicking nature. A school roof gets very hot. Which animal or plant could give you an idea? Explain your design.** [2]

e.g. like a termite mound: tall chimneys/vents let hot air escape; or like a tree: a roof garden or shade leaves; light colours like reflective shells.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

INVESTIGATE · A7 · ENVIRONMENTS

Comparing two habitats

★★☆
Grade 5

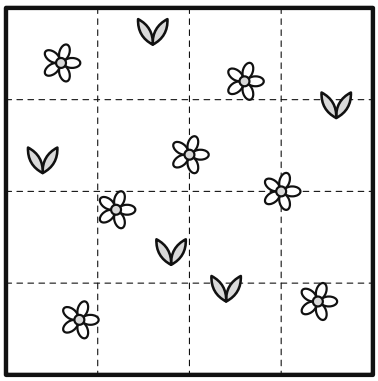
Learning goals

- Use a quadrat to sample plants fairly.
- Compare biodiversity using data.
- Explain differences between habitats.

Class 5A compared two habitats in the school grounds: the **football field** and the **garden by the pond**. In each habitat they threw a 1 m × 1 m **quadrat** at random three times and counted the different kinds of plants and small animals inside.

Remember

Throwing the quadrat at random makes the test fair: you do not choose only the interesting spots.

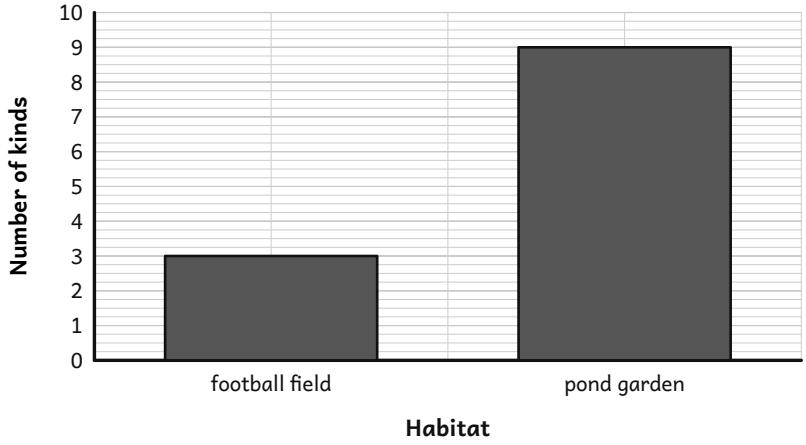


1 metre × 1 metre quadrat

1 **Here are the results. Work out the totals.** [2]

Kind of living thing	Football field	Pond garden
grass	✔	✔
clover	✔	✔
dandelion		✔
fern		✔
moss		✔
ant	✔	✔
earthworm		✔
snail		✔
spider		✔
Number of different kinds	3	9

2 **Draw a bar chart comparing the number of different kinds of living things.** [3]



3 **Which habitat has greater biodiversity? Suggest two reasons.** [3]

The pond garden: it has water, shade, different plants and is not trampled or mowed, so there are more kinds of food and shelter.

4 **Why did the class use the quadrat three times in each habitat?** [1]

One sample might not be typical; repeating gives more reliable results.

5 **Suggest one way the school could increase biodiversity on the field.** [1]

e.g. leave a wild strip unmowed, plant native flowers and trees, make a small pond or log pile.

ANSWER KEY

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LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · A7 · ENVIRONMENTS

The Sundarbans: a forest that protects us

★★★ Grade 5

Learning goals

- Explain why an ecosystem matters to people.
- Evaluate the effect of a human activity, using evidence.
- Argue for a way to reduce harm.

Science · Geography · Economics · Ethics · Climate change

The **Sundarbans** is the largest mangrove forest in the world, about 10 000 km², shared by Bangladesh (about 60%) and India. Mangrove trees such as *sundari* and *keora* grow in salty, muddy water and have special roots that poke up out of the mud to breathe.

The forest is home to Royal Bengal tigers, spotted deer, crocodiles, river dolphins and hundreds of kinds of birds and fish. Hundreds of thousands of people depend on it for fish, crabs, honey and wood.

The Sundarbans also **protects** people. When cyclones such as **Sidr** (2007) and **Aila** (2009) struck, the mangroves slowed the wind and the storm surge, protecting villages behind them.

But the forest faces threats. In **December 2014**, an oil tanker sank in the Shela River inside the forest, spilling about 350 000 litres of heavy oil. Oil coated mud, roots and animals. Rising sea levels, more salt in the water, shipping, cutting of trees and poaching are other dangers.

Sources: UNESCO World Heritage listing; UNEP/UN assessment of the 2014 Shela River oil spill.

1 **Give three ways the Sundarbans helps people.** [3]

It protects villages from cyclones and storm surges; gives fish, crabs, honey and wood for jobs and food; supports tourism; stores carbon.

2 **Explain how an oil spill could harm the food web, starting with the producers.** [3]

Oil coats leaves, roots and algae so producers cannot get light/air and may die; animals that eat them lose food; oil also poisons fish and crabs, so predators are affected.

3 **Different people want different things from the river. Complete the table.** [3]

Person	What they want
a ship owner	a short, cheap route through the forest rivers
a fisher	clean water so there are plenty of fish and crabs
a conservation scientist	to protect tigers, dolphins and the mangroves

4 **Argue from evidence. Should cargo ships be allowed to travel through the Sundarbans? Give your opinion, one piece of evidence and one way to reduce harm.** [4]

Accept a reasoned opinion, e.g. "No, or only with strict rules: the 2014 spill harmed the forest. Ships could use another route, be double-hulled, travel slowly and carry spill-cleaning equipment."

ANSWER KEY

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REVIEW AND CHALLENGE · A7 · ENVIRONMENTS

Show what you know

★★★
Grade 5

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

M	E	V	P	R	O	D	U	C	E	R	I
D	L	E	N	E	R	G	Y	U	M	L	P
D	E	D	P	O	P	P	J	C	A	E	D
K	I	C	P	Y	F	R	A	G	N	R	L
E	Y	C	O	S	A	R	J	V	G	C	Y
I	R	L	O	M	H	F	L	H	R	S	S
R	K	V	H	N	P	A	U	G	O	H	M
H	T	G	R	P	S	O	B	L	V	A	A
I	I	P	I	G	Y	U	S	I	E	U	L
O	G	L	P	R	E	Y	M	E	T	L	C
H	E	D	H	P	G	K	G	E	R	A	P
U	R	P	R	E	D	A	T	O	R	U	T

consumer **decomposer** **energy** **habitat** **mangrove** **predator** **prey** **producer** **tiger**

2 **In the chain grass → deer → tiger, the deer is:** [1]

- ○ a producer
- ● **a herbivore**
- ○ a carnivore
- ○ a decomposer

3 **Which is an example of biomimicry?** [1]

- ○ planting trees
- ● **Velcro copied from burrs**
- ○ a food chain
- ○ a national park

4 **Draw a food web with at least five living things from a pond. Use arrows correctly.** [4]

My pond food web

e.g. algae → tadpole → fish → heron; pond weed → snail → duck; algae → water flea → fish. Arrows from food to eater.

5 **Challenge. Explain why matter is recycled in an ecosystem but energy is not.** [2]

Decomposers break dead things into nutrients that plants use again, so matter goes round. Energy is used up/given off as heat at each step and must keep coming in from the Sun.